DATA ANALYTICS FOR BUSINESS

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Discrete Choice Models

Learning Objectives

- Describe the discrete choice model framework
- Set up a multinomial Logit model
- Articulate identification conditions of discrete choice models
- Estimate discrete choice models
- Calculate and interpret the marginal effects for discrete choice models

Discrete Choice

- Discrete Choice
  - **Binary Response**: Choose between two alternatives
  - **Discrete Choice:** Choose among *multiple* alternatives
- Examples:
  - Choice of band for the same type of products
  - Choice of transportation mode
  - Choice of college/program
  - ……
- The Choice Set
  - Mutually exclusive
  - Exhaustive
  - Finite

Discrete Choice Models

- **Random Utility Model (RUM)**
  - Individual i's utility of choosing alternative j from n alternatives:

$$U_{ij} = V_{ij} + \varepsilon_{ij}$$

  - $V_{ij} = V(X_{ij}; \beta)$: *representative utility* (the observed characteristics affecting the utility)
  - ε_{ij}: random error term (the *unobserved* characteristics affecting the utility)

- **Choice Probability**
  - The probability of individual i choosing alternative j (over all other alternatives)

$$\begin{aligned} P_{ij} &= \Pr(U_{ij} > U_{ik}, \forall k \neq j) \\ &= \Pr(V_{ij} + \varepsilon_{ij} > V_{ik} + \varepsilon_{ik}, \forall k \neq j) \\ &= \Pr(\varepsilon_{ik} - \varepsilon_{ij} < V_{ij} - V_{ik}, \forall k \neq j) \end{aligned}$$

Multinomial Logit Model (1/4)

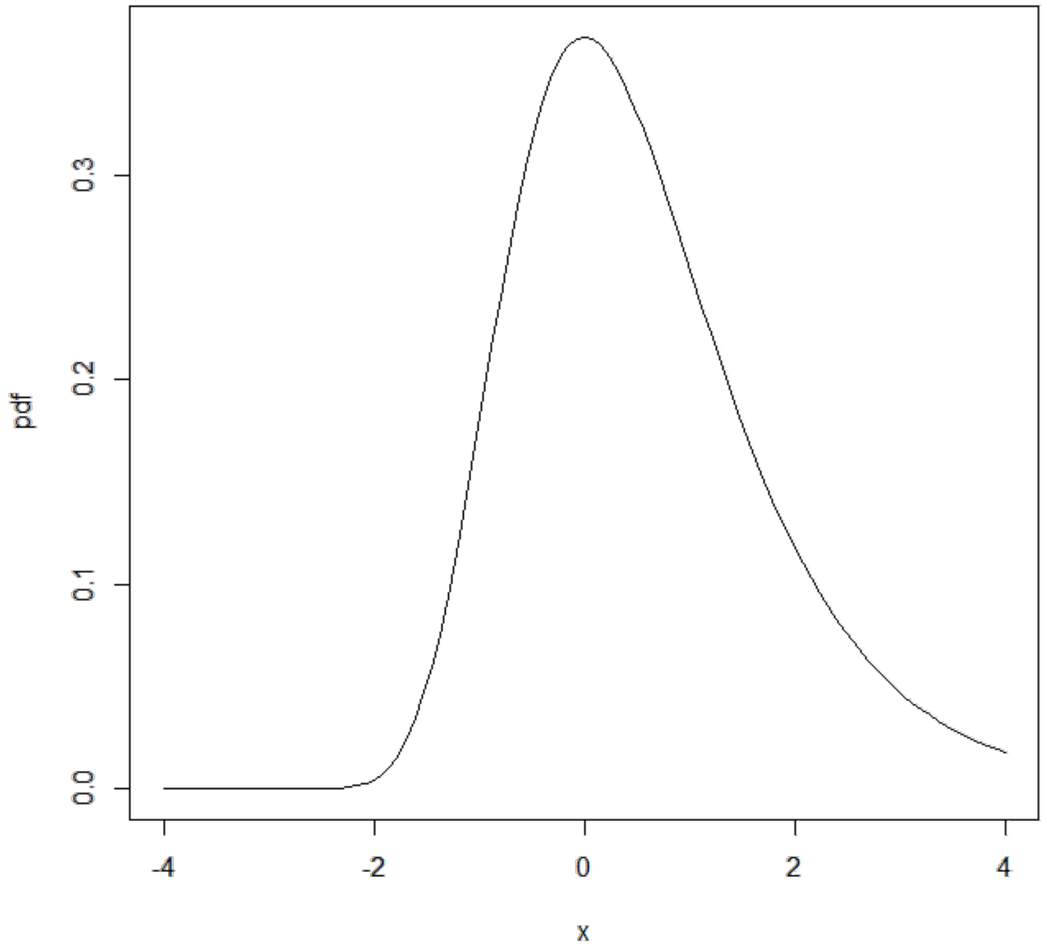
- Choice Probability depends on the distribution of ε
  - Logit: extreme value distribution
  - Probit: multivariate normal distribution
  - Nested Logit, Mixed Logit, …

- Multinomial Logit:
  - $\varepsilon_{ij} \sim$ i.i.d. extreme value distribution for all alternatives j

- <u>Type I (Gumbel) Extreme Value Distribution</u>:

$$f(\varepsilon) = e^{-\varepsilon} e^{-e^{-\varepsilon}}$$

$$F(\varepsilon) = e^{-e^{-\varepsilon}}$$



Multinomial Logit Model (2/4)

- $E(\varepsilon) = \gamma$: the Euler's constant (≈ 0.5772) – Not zero!
- $Var(\varepsilon) = \frac{\pi^2}{6}$

- Empirically indistinguishable from normal
- Slightly skewed (not symmetric about zero)
- Slightly fatter tails

- Desirable analytical properties – closed form!

Multinomial Logit Model (3/4)

- Choice Probability:

$$
\begin{aligned}
P_{ij} &= \Pr(V_{ij} + \varepsilon_{ij} > V_{ik} + \varepsilon_{ik}, \forall k \neq j) \\
&= \Pr(\varepsilon_{ik} < \varepsilon_{ij} + V_{ij} - V_{ik}, \forall k \neq j) \\
&= \prod_{k \neq j} \Pr(\varepsilon_{ik} < \varepsilon_{ij} + V_{ij} - V_{ik}) \\
&= \int \left(\prod_{k \neq j} e^{-e^{-(\varepsilon_{ij} + V_{ij} - V_{ik})}} \right) e^{-\varepsilon_{ij}} e^{-e^{-\varepsilon_{ij}}} d\varepsilon_{ij} \\
&= \cdots \\
&= \frac{\boldsymbol{e^{V_{ij}}}}{\sum_{\boldsymbol{k}} \boldsymbol{e^{V_{ik}}}}
\end{aligned}
$$

Multinomial Logit Model (4/4)

- Binary logit can be viewed as a <u>special case</u>:
  - $V_{i0} = 0, V_{i1} = V$

$$\Pr(y_i = 1) = \frac{e^V}{1 + e^V}$$

  - The *difference* between two extreme value variables $\varepsilon^* = \varepsilon_j - \varepsilon_k$ follows the ***logistic*** distribution

$$F(\varepsilon^*) = \frac{e^{\varepsilon^*}}{1 + e^{\varepsilon^*}}$$

Model Identification

- Only *Differences* in Utility Matter!
  - Alternative-specific constants
    - One of them needs to be normalized to zero!
  - Individual-specific demographic variables
    - Must have alternative-specific coefficient, and one needs to be normalized to zero!

- Overall *Scale* of Utility is Irrelevant!
  - Variance of the error term needs to be normalized
    - Automatically taken care of by the extreme value distribution

Model Specification – General

- General model specification

$$V_{ij} = \alpha_j + \beta x_{ij} + \gamma_j z_i + \delta_j w_{ij}$$

  - α_j: alternative-specific intercepts
  - β: generic coefficients for alternative-specific variables x_{ij}
  - γ_j: alternative-specific coefficients for individual-specific variables z_i
  - δ_j: alternative-specific coefficients for alternative-specific variables w_{ij}

Model Specification – Basic

- Basic model:

$$V_{ij} = X_{ij}\beta_j = \beta_{0j} + \beta x_{ij}$$

  - β: generic coefficients for alternative-specific variables x_{ij}
    - x_{ij}: attributes specific to each alternative
  - β_{0j}: alternative-specific intercepts
    - One needs to be normalized to zero
    - If J alternatives in total, only $J - 1$ intercepts

Model Estimation

- Likelihood function:

$$L(\beta) = \prod_{i=1}^{n} \prod_{j=1}^{J} (P_{ij})^{y_{ij}} = \prod_{i=1}^{n} \prod_{j=1}^{J} \left(\frac{e^{X_{ij}\beta_j}}{\sum_k e^{X_{ik}\beta_k}} \right)^{y_{ij}} = \prod_{i=1}^{n} \frac{e^{X_{ij_i}\beta_{j_i}}}{\sum_k e^{X_{ik}\beta_k}}$$

  - $y_{ij} = 1$ if individual i chooses alternative j; $y_{ij} = 0$ otherwise
  - j_i denotes the alternative individual i chose

- MLE estimation
  - Log-likelihood function globally concave in β

Marginal Effects

- Marginal Effects on the Choice Probability of the *Same* Alternative:

$$
\begin{aligned}
\boldsymbol{\frac{\partial P_{ij}}{\partial x_{ij}}} &= \frac{\partial}{\partial x_{ij}}\left(\frac{e^{V_{ij}}}{\sum_k e^{V_{ik}}}\right) \\
&= \frac{e^{V_{ij}}}{\sum_k e^{V_{ik}}}\frac{\partial V_{ij}}{\partial x_{ij}} - \frac{e^{V_{ij}}}{(\sum_k e^{V_{ik}})^2} e^{V_{ij}} \frac{\partial V_{ij}}{\partial x_{ij}} \\
&= \frac{\partial V_{ij}}{\partial x_{ij}}\left(P_{ij} - P_{ij}^2\right) = \boldsymbol{\beta_{xj} P_{ij}\left(1 - P_{ij}\right)}
\end{aligned}
$$

- Note that the marginal effect is the greatest when $P_{ij} = \frac{1}{2}$.

Marginal Effects (Cont'd)

- Marginal Effects on the Choice Probability of *Other* Alternatives:

$$\frac{\partial P_{ij'}}{\partial x_{ij}} = \frac{\partial}{\partial x_{ij}}\left(\frac{e^{V_{ij'}}}{\sum_k e^{V_{ik}}}\right)$$
$$= -\frac{e^{V_{ij'}}}{(\sum_k e^{V_{ik}})^2} e^{V_{ij}} \frac{\partial V_{ij}}{\partial x_{ij}}$$
$$= -\frac{\partial V_{ij}}{\partial x_{ij}} P_{ij} P_{ij'} = -\beta_{xj} P_{ij} P_{ij'}$$

- Note that changes in all choice probabilities caused by one variable add up to zero:

$$\sum_{k=1}^{J} \frac{\partial P_{ik}}{\partial x_{ij}} = 0$$

Summary

- Discrete choice model framework
- Multinomial Logit model setup
- Identification conditions of discrete choice models
- Discrete choice model estimation
- Marginal effects for discrete choice models